In the last lecture

Conditional Construct

- if
- if-else

Loop - Repetition structure

Example

```
int sum ;
sum = 1+2+3+4+5+.......+10 ;
cout << sum ;
```

Find the Sum of the first 100 Integer starting from 1

?

while

```
while ( Logical Expression )
{
 statements;     :
}
```

```
int sum ;
sum = 0 ;
```

int sum = 0; (Optional)

Example

```
int sum , number ;
sum = 0 ;
number = 1 ;
while ( number <= 1000 )
{
  sum  =  sum  +  number ;
    number  =  number + 1 ;
}
cout << " The sum of the first 1000 integer starting from 1 is " << sum ;
```

```
while (number <= UpperLimit)
```

Example

```
int sum, number , UpperLimit ;
sum = 0 ;
number = 1 ;
cout << " Please enter the upper limit for which you want the sum " ;
cin >> UpperLimi t;
while (number <= UpperLimit)
{
  sum = sum + number ;
  number = number +1 ;
}
cout << " The sum of the first " << UpperLimit << " integer is " << sum ;
```

```
if ( number % 2 == 0 )
{
  sum = sum + number ;
  number = number + 1 ;
}
```

Example

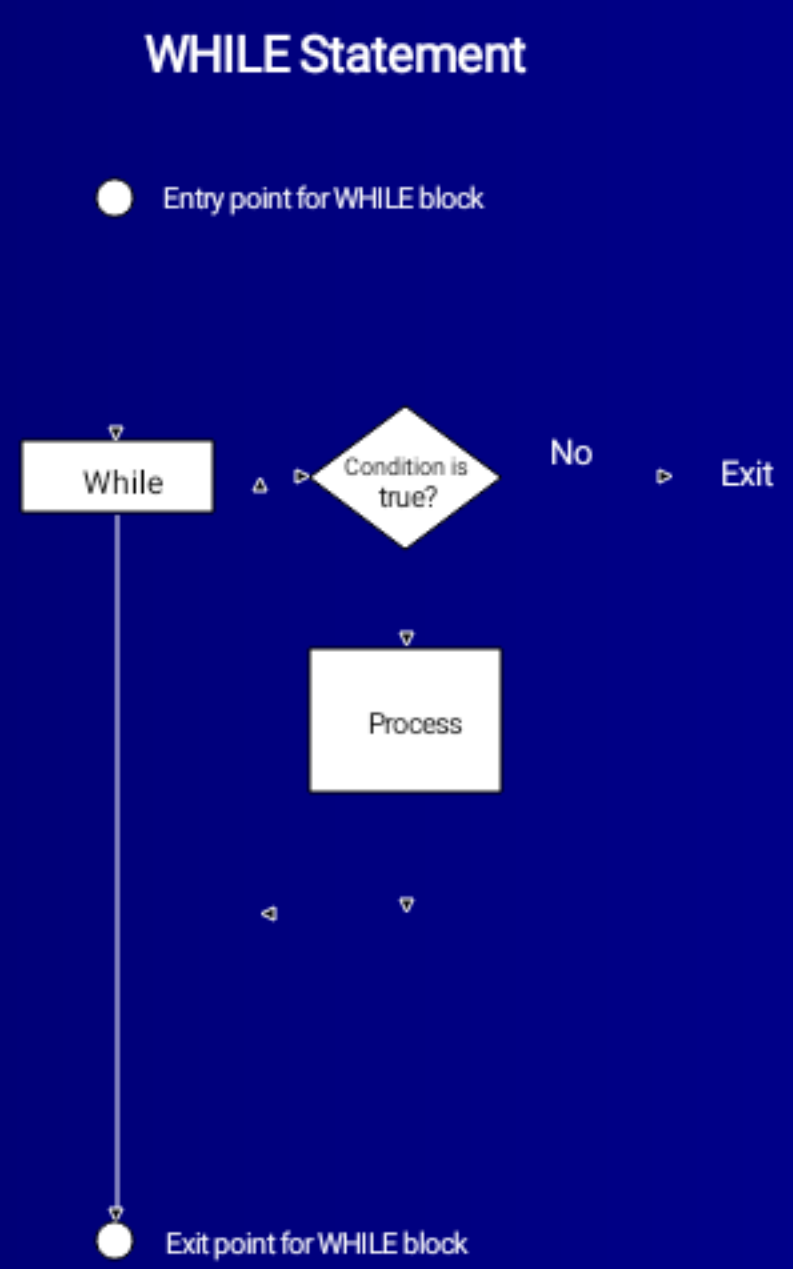
```
sum = 0;
number = 1;
cout << " Please enter the upper limit for which you want the sum ";
cin >> UpperLimit;
while (number <= UpperLimit)
{
 if (number % 2 == 0)
 {
  sum = sum + number;
  number = number + 1;
 }
}
cout << " The sum of all even integer between 1 and " << UpperLimit << " is" << sum;
```

```
2 * ( number / 2 ) ;
```

?

```
int Junk ;
Junk = 1 ;
while ( Junk <= UpperLimit )        ( infinite loop ) X
{
  sum = sum + number ;
  number = number + 1 ;
}
```

Flow Chart for While Construct



Factorial Definition

$n! = n*(n-1)*(n-2)*(n-3)\ldots\ldots\ldots\ldots*3*2*1$

Example: Factorial

```
#include <iostream.h>
main ( )
{
 int number ;
 int factorial ;
 factorial = 1 ;
 cout << “Enter the number of Factorial” ;
 cin >> number ;
 while ( number >= 1 )
  {
   factorial = factorial * number ;
   number = number – 1 ;
  }
 cout << “Factorial is” << factorial ;
}
```

Property of While Statement

It executes zero or more times